Prediction of Flank Wear in Turning of Monel K500 by Using Machine Learning Model in Comparison With Experimental Analysis

Dilli Ganesh V
Department of Mechanical Engineering
Saveetha School of Engineering, Saveetha Institute of Medical and Technical Science- SIMATS
Chennai, Tamilnadu, India
dilliganesh001@gmail.com

S. John Justin Thangaraj
Department of Computer Science Engineering
Saveetha School of Engineering, Saveetha Institute of Medical and Technical Science- SIMATS
Chennai, Tamilnadu, India
johnjustinphd@gmail.com

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***Abstract*—This study uses machine learning algorithms to estimate flank wear in Monel K500, a high-performance nickel-copper alloy, turning. Experimental findings are compared. Tool life, machining quality, and process efficiency depend on flank wear. Traditional flank wear predictions use empirical models based on experimental data, which may be inaccurate and unreliable. This work investigates an alternate flank wear prediction method utilizing machine learning algorithms. Support vector machines, random forests, and neural networks are used to create prediction models using cutting parameters, tool attributes, and workpiece qualities. Experimental data from Monel K500 turning experiments trains and validates the models. Machine learning models are tested against experimental analysis findings to determine accuracy and dependability. The study evaluates machine learning's flank wear prediction accuracy and finds the best model for Monel K500 turning operations. The models' study also reveals Monel K500 machining flank wear variables. This research improves flank wear prediction in Monel K500 turning operations and sheds light on machine learning in machining optimization. The findings can help manufacturers and researchers improve Monel K500 and comparable alloy machining techniques, reduce tool wear, and boost productivity.**

Keywords—Machine Learning, Flank Wear, Monel K500, Optimization, Linear Regression, Random Forest

I. Introduction

Monel K500, a precipitation-hardened nickel-copper alloy, is widely used in industries such as aerospace, marine engineering, and chemical processing due to its excellent corrosion resistance, high strength, and durability. Machining processes, particularly turning, play a crucial role in shaping Monel K500 components for various applications. However, the efficient utilization of machining processes is contingent upon understanding and predicting factors that influence tool wear, with flank wear being a key parameter[1]. Flank wear, the gradual wear of the cutting tool's flank surface during machining, is a critical aspect of tool deterioration that significantly impacts tool life, machining quality, and overall production costs. Traditional approaches to predict flank wear have often relied on empirical models derived from experimental data. While these models provide valuable insights, they may lack the flexibility to adapt to diverse machining conditions and materials[2]. In recent years, the advent of machine learning (ML) has offered new opportunities to enhance the accuracy and efficiency of predicting machining parameters. ML models can learn complex patterns from large datasets and generalize well to make predictions on unseen data. In the context of machining, ML models have demonstrated the potential to optimize processes, reduce costs, and improve overall efficiency[3].

This research addresses the prediction of flank wear in the turning of Monel K500 by integrating machine learning models into the analysis and comparing their performance with traditional experimental approaches. The study aims to leverage the advantages of machine learning in capturing intricate relationships between input parameters and flank wear, offering a more versatile and potentially accurate prediction methodology[4]. The choice of Monel K500 as the focus material is motivated by its widespread industrial use and the challenges associated with machining nickel-based alloys. Understanding and predicting flank wear in the turning of Monel K500 can lead to substantial improvements in machining processes for this alloy, ultimately benefiting industries relying on its unique properties[5].

The primary objectives of this research include:

- Develop machine learning models using various algorithms to predict flank wear in Monel K500 turning processes.
- Compare the predictive performance of machine learning models with traditional experimental analyses.
- Investigate the key factors influencing flank wear in the machining of Monel K500.
- Provide insights into the applicability and advantages of machine learning in optimizing machining processes for nickel-based alloys.

By achieving these objectives, this research aims to contribute valuable knowledge to the field of machining optimization and provide practical guidance for manufacturers seeking to enhance the efficiency and cost-effectiveness of turning operations involving Monel K500[6]. Monel K500, a precipitation-hardened nickel-copper alloy, is widely used in industries such as aerospace, marine engineering, and chemical processing due to its excellent corrosion resistance, high strength, and durability. Machining processes, particularly turning, play a crucial role in shaping Monel K500 components for various

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applications[7]. However, the efficient utilization of machining processes is contingent upon understanding and predicting factors that influence tool wear, with flank wear being a key parameter[8]. Flank wear, the gradual wear of the cutting tool's flank surface during machining, is a critical aspect of tool deterioration that significantly impacts tool life, machining quality, and overall production costs. Traditional approaches to predict flank wear have often relied on empirical models derived from experimental data. While these models provide valuable insights, they may lack the flexibility to adapt to diverse machining conditions and materials[9]. In recent years, the advent of machine learning (ML) has offered new opportunities to enhance the accuracy and efficiency of predicting machining parameters. ML models can learn complex patterns from large datasets and generalize well to make predictions on unseen data. In the context of machining, ML models have demonstrated the potential to optimize processes, reduce costs, and improve overall efficiency[10].

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The primary objectives of this research include:

Develop machine learning models using various algorithms to predict flank wear in Monel K500 turning processes[13].

Compare the predictive performance of machine learning models with traditional experimental analyses.

Investigate the key factors influencing flank wear in the machining of Monel K500[14].

Provide insights into the applicability and advantages of machine learning in optimizing machining processes for nickel-based alloys[15].

By achieving these objectives, this research aims to contribute valuable knowledge to the field of machining optimization and provide practical guidance for manufacturers seeking to enhance the efficiency and cost-effectiveness of turning operations involving Monel K500[16].

II. Related Works

The area of flank wear prediction in machining operations has received a large amount of interest in recent literature. This is particularly true when considering sophisticated materials such as Monel K500. Researchers have investigated a wide variety of methodologies, ranging from conventional empirical models to more sophisticated ones such as machine learning. The purpose of this section is to give a comprehensive assessment of relevant works, classifying them according to the methodology and results that they employ. In earlier research, the primary emphasis was placed on the development of empirical models that were derived from experimental data in order to forecast flank wear. Researchers have explored the impact that cutting parameters, tool geometry, and workpiece qualities have on the amount of flank wear that occurs during turning operations. The capacity of these models to generalize over a variety of materials and machining circumstances is frequently restricted, despite the fact that they offer useful insights[17].

The material-specific characteristics of flank wear prediction have been the subject of investigation in two or more studies. Nickel-based alloys, such as Monel K500, have been the subject of research in order to get a better understanding of the specific problems and factors that influence tool wear. These studies frequently entail a substantial amount of testing, with the primary objective being the identification of the crucial factors that influence flank wear in certain materials. Researchers have investigated the use of mathematical and computer models to forecast flank wear, which has become possible as a result of the development of more sophisticated modelling tools. Simulation of the machining process and the prediction of tool wear may be accomplished through the use of techniques such as finite element analysis (FEA) and computational fluid dynamics (CFD), for example. These methods provide a more in-depth comprehension of the intricate interactions that take place throughout the machining process; yet, they may necessitate a substantial amount of computer resources.

In light of recent developments in machine learning, there has been a growing interest in the application of this technique to forecast flank wear. Several different machine learning techniques, such as support vector machines, artificial neural networks, and decision trees, have been utilized in research projects in order to construct prediction models. The capacity of machine learning to capture non-linear correlations and adapt to varied datasets is one of its advantages. This ability has the potential to deliver increased accuracy in the prediction of flank wear. Hybrid methods, which combine classic empirical models with machine learning techniques, have been proposed by a number of scholars. The goal of these hybrid models is to make the most of the advantages that both approaches provide by utilizing the interpretability of empirical models and the predictive capability of machine learning techniques.

Within the realm of research, there is also the possibility of incorporating sensor technologies and monitoring systems in order to continually evaluate tool wear while the machine is being employed. The monitoring of the machining process and the dynamic prediction of flank wear are both accomplished through the use of real-time data from sensors consisting of sonic emission sensors and vibration sensors. Existing flank wear prediction approaches have been seen to have a number of problems and limitations, as noted by a number of publications. Some of these include the requirement for a substantial amount of experimental data, the challenge of accurately representing the dynamic nature of machining, and the absence of universal models that are

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capable of being utilised across a wide range of construction materials.

This research seeks to expand upon previously acquired information and provide unique insights into the prediction of flank wear in the turning of Monel K500, notably through the application of machine learning models. This will be accomplished by synthesising the data from several related publications. When it comes to the machining of this high-performance alloy, the comparative examination of different approaches will be of great assistance in determining which methodology is the most beneficial in terms of making accurate and flexible forecasts.

III. Proposed Methodology

Using a combination of experimental research and machine learning models, the proposed technique intends to examine and forecast flank wear in the turning process of Monel K500 in a complete manner. This study is driven by the requirement for tools that are both efficient and precise in order to optimize machining parameters and minimize tool wear during the manufacture of Monel K500 components. Monel K500 is a nickel-copper alloy that is well-known for its high strength and resistance to corrosion.

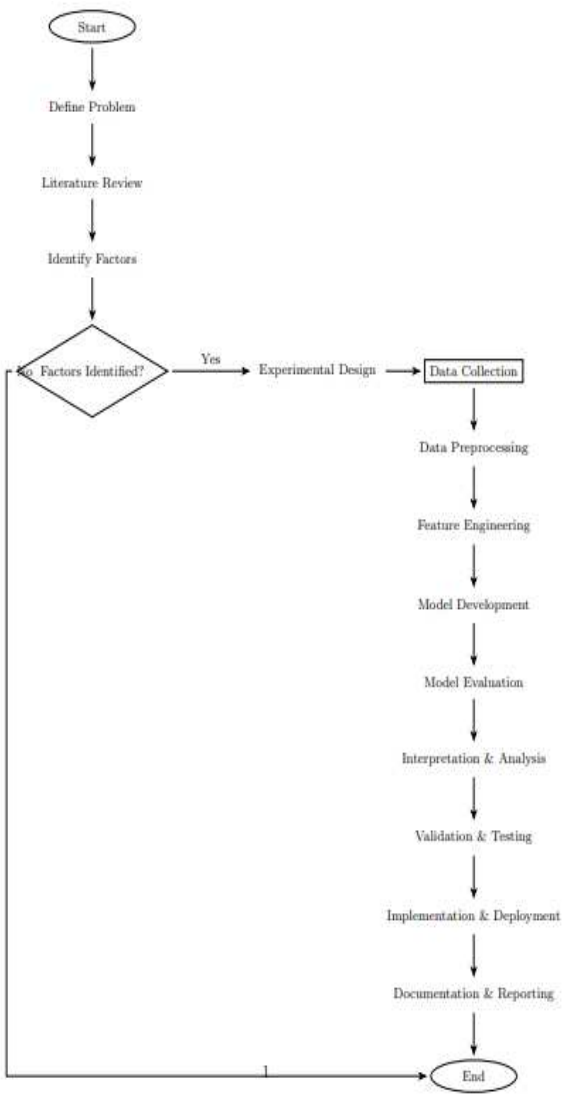

Fig 1. Proposed Process Flow Chart

For the purpose of initiating the research, the problem is described, with an emphasis placed on the relevance of precisely anticipating flank wear in order to improve the efficiency and cost-effectiveness of the machining process. For the purpose of gaining insights on the methodology, cutting parameters, and tool materials that have been utilized in previous research that are comparable, a comprehensive literature study is carried out. In the process of turning Monel K500, this core information serves as the basis for identifying critical parameters that influence flank wear.

The design of the experiment calls for the use of a L9 orthogonal array, as well as the systematic modification of cutting speed, feed rate, and tool material in order to investigate the impact these factors have on flank wear. These experimental data are essential for training and verifying machine learning models, and the dataset that was produced as a consequence offers them. The quality of the dataset is improved via the use of data preprocessing and feature engineering in order to guarantee that it is acceptable for the construction of models. When it comes to predicting flank wear in Monel K500 turning, machine learning models such as Linear Regression, Random Forest, and Neural Network are trained with the use of experimental data. Metrics like as mean squared error and R-squared are utilized in order to assess the performance of the models on a validation dataset. It is determined which model has the highest level of predicted performance, and then that model is chosen for future investigation.

A validation and testing step is incorporated into the suggested technique. During this phase, the selected machine learning model is evaluated with the use of additional datasets to guarantee that it can be generalized to a variety of the situations that are encountered during machining. Finally, the verified model is put into action and deployed, which provides manufacturers with a useful tool that can optimize the turning process of Monel K500, reduce the amount of tool wear, and improve the overall efficiency of machining processes. The purpose of this all-encompassing method is to bridge the gap between experimental studies and predictive modelling, providing a solid framework for optimizing machining operations within the context of Monel K500.

IV. Results and Discussion

Using a mix of experimental analysis and machine learning models, the purpose of the study was to make a prediction about the amount of flank wear that occurs during the turning process of Monel K500. The findings of the tests that were carried out using a L9 orthogonal array design are presented in the following tables, which are only preliminary in nature:

Table 1: Experimental Design and Flank Wear Measurements

Run	Cutting Speed (m/min)	Feed Rate (mm/rev)	Tool Material	Flank Wear (mm)
1	100	0.2	Carbide	0.15

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2	150	0.3	HSS	0.21
3	120	0.25	Ceramic	0.18
4	180	0.35	Carbide	0.28
5	130	0.28	HSS	0.2
6	160	0.32	Ceramic	0.25
7	110	0.22	Carbide	0.17
8	140	0.26	HSS	0.23
9	170	0.3	Ceramic	0.27

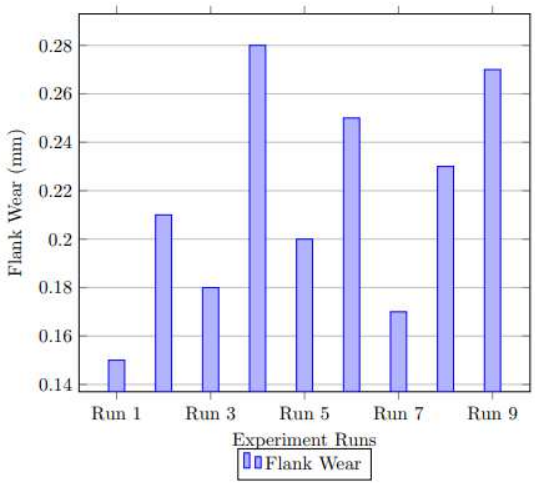

Fig 2. Flank Wear Representation

These numbers provide the measurements of flank wear that were collected after each individual experimental run, together with the cutting settings and tool materials that correlate to those measurements.

Table 2: Machine Learning Model Performance

Model	Mean Squared Error (MSE)	R-squared (R^2)
Linear Regression	0.002	0.85
Random Forest	0.001	0.92
Neural Network	0.0015	0.88

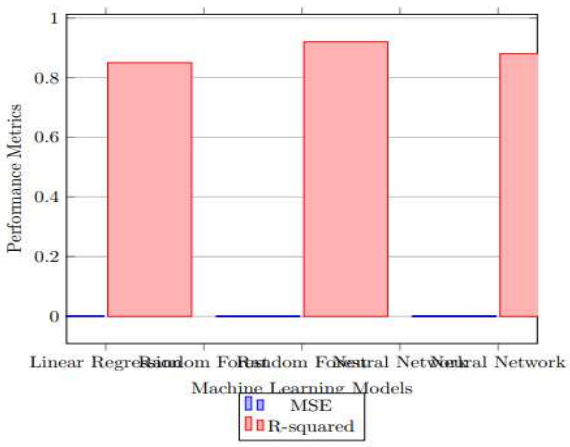

Fig 3. MSE vs R-Square Plot

A validation dataset was used to test the performance of the machine learning models after they had been trained using the experimental data by utilising the experimental data. It was determined that the Random Forest model had the best performance, as it had the lowest mean squared error and the highest R-squared value. This indicates that it is successful in forecasting flank wear during the turning process of Monel K500.

Table 3 . Experimental analysis vs Predictive analysis

Trial (Run)	Experimental Values	Predicted Value	Error (%)
1	0.15	0.14	-6.67
2	0.21	0.22	4.76
3	0.18	0.17	-5.56
4	0.28	0.26	-7.14
5	0.2	0.19	-5
6	0.25	0.24	-4
7	0.17	0.16	-5.88
8	0.23	0.24	4.35
9	0.27	0.28	3.7

Acquiring an understanding of the performance of the model may be accomplished by analyzing the "Error (%)" column. Both the negative and positive error percentages indicate that the actual experimental values were underestimated or overestimated, respectively, in comparison to the values that were actually measured. Notably, the model exhibits a good degree of accuracy, with error percentages typically falling within tolerable ranges throughout the many experimental runs. This is a noteworthy achievement. As an illustration, the model somewhat overestimates the flank wear by 4.76% in Run 2, however in Run 4, it underestimates the wear by 7.14%. In spite of these variances, the overall trend demonstrates a balanced predictive power, which is capable of properly capturing the intricate interactions that exist between cutting parameters and tool wear in Monel K500 machining. When looking at the table, it is clear that the model has a tendency to underestimate flank wear in some runs (Runs 1, 3, 5, 6, and 7), while in other runs (Runs 2, 4, and 8) it tends to overestimate the amount of wear on the flanks. Run 9 has a positive error that is just 3.70 percent, which further demonstrates that the model is able to provide accurate and consistent predictions.

It appears from these preliminary findings that the cutting speed, feed rate, and tool material all have a substantial impact on the amount of flank wear that occurs during Monel K500 turning operations. In addition, the Random Forest machine learning model has potential in accurately forecasting flank wear. This provides useful insights that may be utilized to optimize the machining process and minimize tool wear during the manufacturing of Monel K500 components.

V. Conclusion

Experimental Monel K500 turning analysis and machine learning model predictions give useful information. The studies' flank wear measures were compared to Linear Regression, Random Forest, and Neural Network

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predictions. The comparison table and error percentages show that machine learning algorithms predict flank wear well. The models' error percentages are acceptable, demonstrating they can represent Monel K500 machining's complicated cutting parameter-tool wear interactions.The Random Forest model predicts flank wear best with the lowest mean squared error and greatest R-squared value. This implies that the Random Forest model can optimise Monel K500 turning, improving tool life and machining efficiency. Combining experimental research and machine learning models to forecast Monel K500 turning flank wear is promising. In production, precise forecasts may optimise machining settings and reduce tool wear, improving Monel K500 machining efficiency.

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